Pilot Study: Batch Reactor Yield Optimization

This short test paper describes a synthetic batch reactor study designed for validating document retrieval and RAG workflows in AI Scientist Copilot. The primary response variable is product yield, reported as yield_pct in the accompanying dataset.

The study found that temperature, residence time, and catalyst loading had the strongest positive association with yield. Pressure showed a mild negative association with yield in the synthetic experiments. Extremely high feed concentration produced unstable impurity behavior.

Recommended operating conditions for a first optimization pass are temperature between 82 and 88 C, residence time between 45 and 55 minutes, catalyst loading between 1.1 and 1.4 wt pct, and pressure below 3.0 bar.

A useful baseline modeling approach is to train a Random Forest regressor using yield_pct as the target column. Important input features are expected to include temperature_C, residence_time_min, catalyst_loading_wt_pct, pressure_bar, and agitation_rpm.

For safe interpretation, missing values should be checked before model training. Rows with missing pressure_bar, agitation_rpm, or feed_concentration_mol_L may require imputation or removal.

Variable	Expected effect on yield
temperature_C	positive up to moderate range
residence_time_min	positive
catalyst_loading_wt_pct	positive
pressure_bar	slightly negative
feed_concentration_mol_L	nonlinear